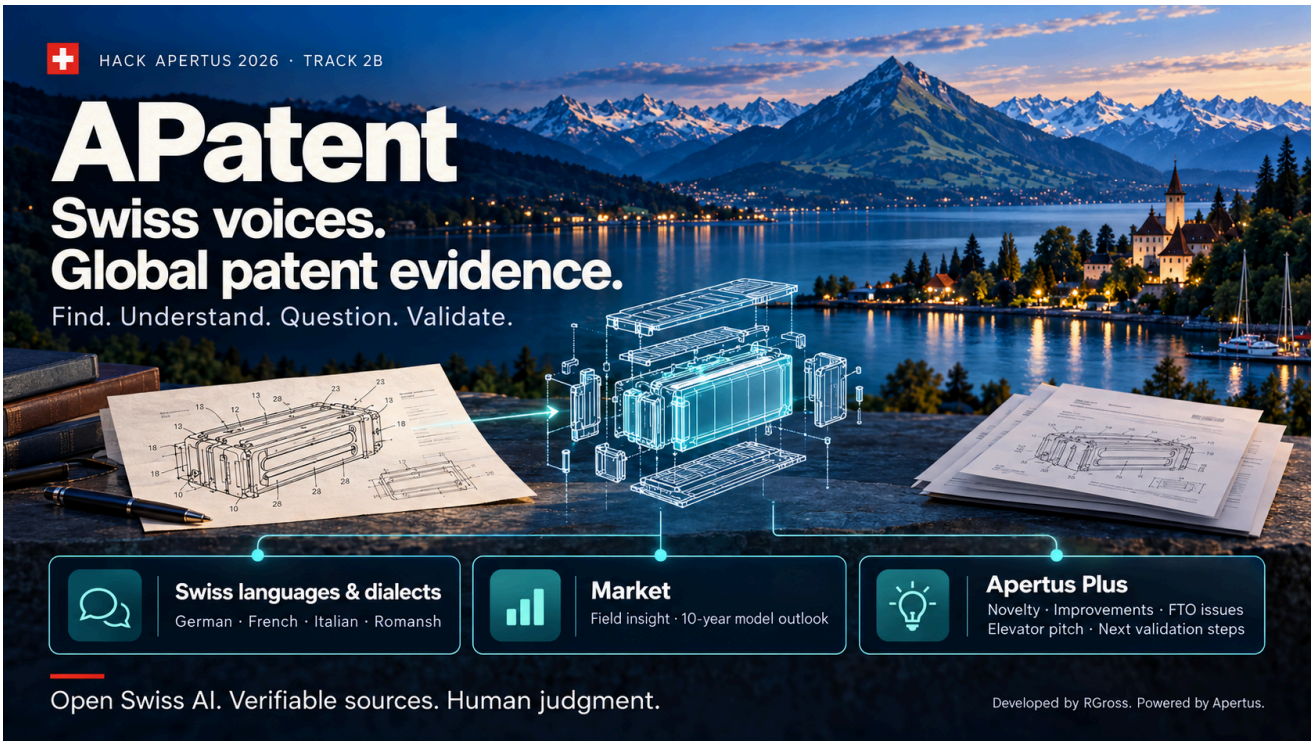
APatent — Technical report

- **Track:** 2B — Apertus Adoption: Own Project
- **Event:** Hack Apertus Online, 1–16 October 2026
- **Team and developer:** RGross
- **Team members:** Rafael Gross and Ricardo Gross
- **Team status:** professionals; not a Swiss student team
- **Maturity:** working prototype with deployed service and automated software checks
- **Demo:** Live application · Video (1m51s)
- **Source:** https://git.rgross.ch/RGross/HackApertus
- **Report date:** 16 October 2026

1. Summary

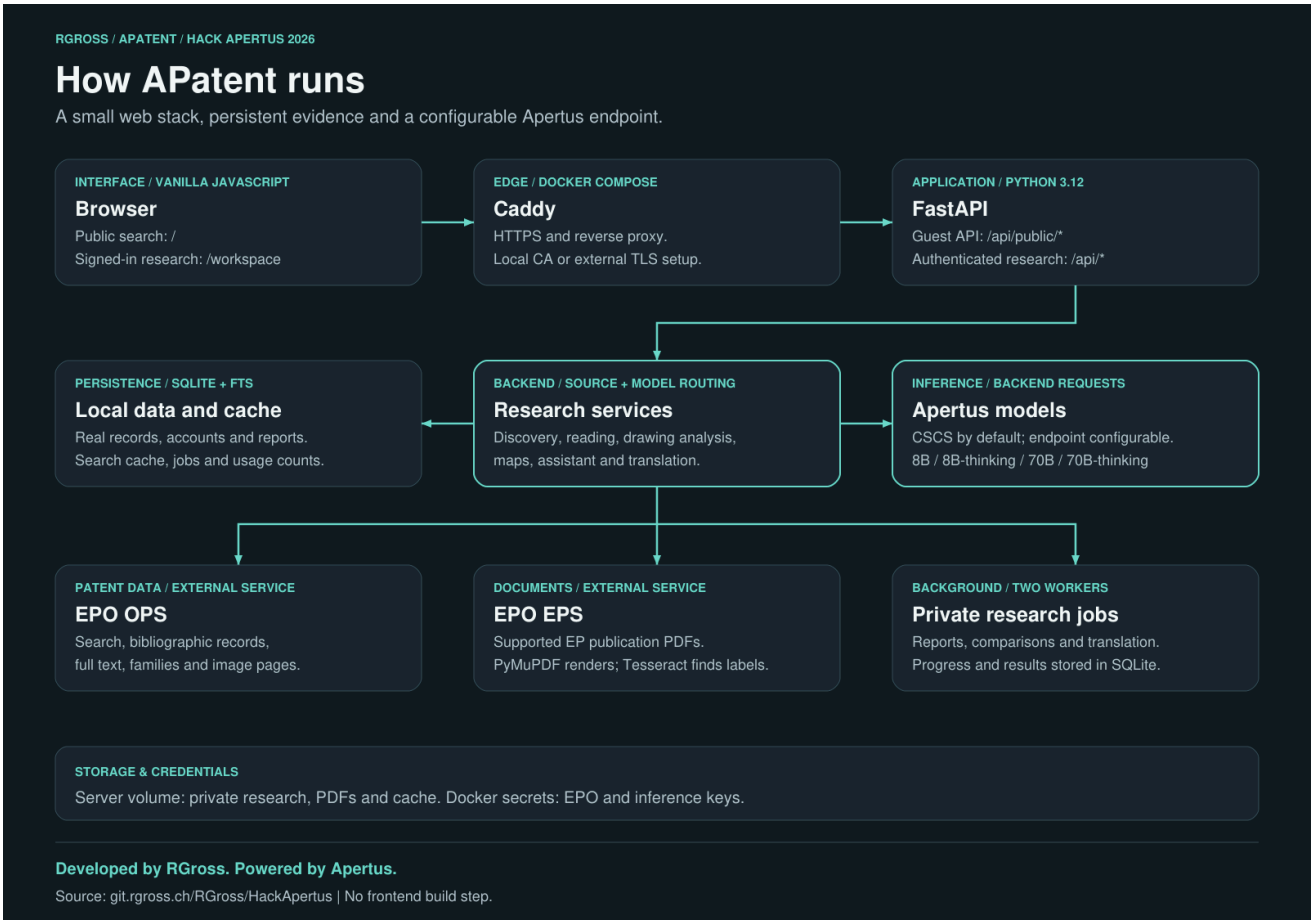


Swiss voices. Global patent evidence. APatent helps inventors and small R&D teams turn real patent records into an explanation in their language and a next validation step. The flow connects EPO search, filing activity, patent reading and drawing inspection. Public search needs no account; private research adds reports, maps, translation and comparison.

Three pillars support that decision: **Swiss languages and dialects** for the summary, Plus and short drawing caption; **Market** for a labeled field estimate and ten-year outlook; **Apertus Plus** for five required assessments and next checks. English source passages remain inspectable. Four Apertus roles plan, explain, explore and challenge the work. Software verification passed 175 Python and 14 JavaScript tests; these establish software behavior, not model accuracy or measured user benefit.

Switzerland ranked first in 2025 EPO applications per person among countries above one million inhabitants. Apertus brings fully open Swiss AI and stated values of transparency, neutrality and accountability. APatent applies that philosophy through visible sources, model routing and human review. [R1–R3]

2. Architecture



The browser uses plain JavaScript. Caddy provides HTTPS. FastAPI handles public and authenticated requests. SQLite stores real records, FTS search, caches, accounts, model routes and research jobs. Two background workers process reports and translations. PyMuPDF renders PDFs; Tesseract locates printed drawing labels. The UI shows source passages separately from model interpretation.

Target architecture: a) on-premise. The application, database, secrets and access policy can run on infrastructure administered by its operator. Inference can be moved to an operator-controlled compatible endpoint. The default CSCS endpoint is an external runtime dependency. Self-hosted inference has not been benchmarked in this project. A complete air-gapped workflow and Swiss cloud data residency have not been established. [R4]

Connection	When	Information sent
EPO OPS	Runtime, when configured	Search terms and filters, publication IDs, image requests
EPO EPS	Runtime for supported EP PDFs	Publication identifier
Configured inference endpoint	Runtime for model features	Bounded patent text, questions, prompts and drawing images
Package/image registries	Build time	Dependency and image requests; no research corpus

Drawing corrections stay in browser storage. Workspace records and private jobs stay in the server database; selected patent evidence and questions go to the model endpoint when the user requests analysis. The web application needs no GPU when inference is remote. Weight serving requires separately sized hardware; no tested local GPU requirement is claimed.

3. Use of Apertus

Default model ID	Work and reason
`swiss-ai/Apertus-v1.5-8B`	Short structured search plans, chart categories and report evidence extraction
`swiss-ai/Apertus-v1.5-8B-thinking`	Application hypotheses and experiments in private reports
`swiss-ai/Apertus-v1.5-70B-thinking`	The short line under a drawing, critical review, and an independent drawing review
`swiss-ai/Apertus-v1.5-70B`	Patent reading, drawing interpretation, maps, translation and final synthesis

This is inference, not training or fine-tuning. The default endpoint is CSCS chat completions; administrators can change approved model routes. Source prompts, validation and request budgets live in `backend/ai.py`, `discovery.py`, `smart_reading.py`, `public_api.py` and the research/translation modules. Temperature is 0.2 for common inference calls. Responses are not assumed deterministic; seeds, quantisation and serving internals are not controlled by this application.

The small model is assigned short structured work. The large models handle longer synthesis and visual interpretation. This is a design choice; a measured 8B-versus-70B quality/cost ablation remains future work. Generated citation IDs are checked against retrieved evidence. That validates references, not every inference.

The page explains what a figure is trying to show and lists only parts it can name. Apertus 70B-thinking adds a short line under the drawing in the chosen language. Without description and claims, a second model reviews the page on its own. The dollar outlook is an unsourced estimate with one growth assumption; it is not market evidence or patent revenue. [R5]

Mandatory Apertus Plus: the reading includes novelty/inventiveness strengths and weaknesses, improvements or alternative embodiments, potential freedom-to-operate issues, a one-sentence elevator pitch and next validation steps. Missing fields fail validation; the UI shows the five assessments in the public page and workspace. These are hypotheses and checks, not a legal clearance.

Development assistance: AI coding assistants supported implementation, documentation and verification under RGross's direction. They are development tools. Application inference uses the configured Apertus roles. No separate automatic model judge, fine-tuned adapter or new model checkpoint is submitted.

4. Data

Live bibliographic records and available patent text come from EPO OPS. Supported EP PDFs come from EPS. The local index stores retrieved or imported real records and has bounded coverage. A fresh installation seeds one real publication, `EP4481893A1`, from the bundled file of about 250 KB. This is not a worldwide corpus.

Tests use bounded XML fixtures and synthetic records for software behavior. There is no newly collected human-subject or training dataset. Public patent records can contain applicant and inventor names; they must not be described as anonymised personal data. User databases, keys and private notes are excluded from the submitted tree. Third-party patent text and drawings retain their rights and service terms. No new dataset licence is granted over them. The submission `data/` directory is below the organizer's 100 MB limit. [R4, R6]

RGross-authored code is offered under Apache-2.0 and documentation under CC-BY-4.0. Dependencies keep their own licences. PyMuPDF/MuPDF is AGPL or commercially licensed; its obligations remain a release review item and are not covered by the project's Apache grant. [R7]

5. Evaluation

The software task is to retrieve scoped evidence, keep unavailable data visible and isolate private research. Automated tests use provider fixtures and controlled model responses. The deterministic source-record path is the baseline; the model layer adds interpretation while keeping that source available.

Setup / check	Result	What it establishes
Python software suite	175 passed	Search, parsing, output validation, auth, ownership and source handling
JavaScript state suite	14 passed	Navigation, page counts, browser notes and normalized overlays
Module syntax / Git whitespace	Passed	Parseable modules and clean patch formatting
No-key source baseline	Bundled real record	Source access without inference; not model performance
Model-quality comparison	Not measured	No accuracy or superiority claim
User benefit / tenfold load	Not measured	No time-saving or throughput claim

Final package startup checks are recorded in `docs/EVALUATION.md`. Historical live evidence is dated there and is not a current health guarantee. A prior 14-page drawing run completed with zero page errors; some component names remained hypotheses. Page coverage and interpretation accuracy are separate measurements. [R5]

Costs depend on tokens and provider rates: `task cost = input tokens × input rate + output tokens × output rate`, using matching billing units. The app records reported tokens and completed/failed calls. No current price or cost per useful task was measured. Public search caches successful OPS results for 600 seconds; planning and reading have additional caches. Cache savings have not been benchmarked.

At ten times the uncached workload, requests and token cost would scale roughly tenfold before capacity effects. This is an assumption, not a load result. Two report workers, SQLite writes, provider limits and per-address limits can add queues or rejection. Next evaluation: measure cold/warm latency, task success, tokens and queue time on a fixed patent set and ten concurrent users.

6. Limitations

OPS can throttle or omit full text and images. The fallback corpus is limited; it never establishes global recall. Annual history shows provider totals or local counts; applicant/geography charts count the displayed page. Current-year data is incomplete. Relevance and citation ordering use a bounded sample; citations mean references inside a record.

Model text can be wrong. Citation validation does not establish legal scope, novelty or commercial value. Translation has not received expert dialect review. A drawing line can still be wrong. Unnamed parts are left out, and a second model review does not remove that limit. Dollar estimates have no measured market baseline.

The final package has no independent user study, quantitative model ablation, verified self-hosted GPU sizing or tenfold load test. The web stack is a working prototype, not a production patent-clearance service. A first external pilot group has not been confirmed.

7. Reproducibility

Install Docker Compose and Make. Clone the complete public repository and run `make run` from its root or `track_2b/`. Open `https://localhost:18443`; trust the local Caddy CA. The HTTP health/source check uses `127.0.0.1:18080`. This jury project has separate volumes from the production deployment. `make stop` stops it without deleting research data.

Set `LLM_BASE_URL`, `LLM_API_KEY` and optionally `LLM_NAME` for inference; use EPO OPS credentials for live search. The default final model is 70B; other Apertus roles remain configured. No keys are included. Without keys, judges can reproduce local search and source inspection, not a successful model call.

For the software suite, create `.venv`, install `requirements.txt`, then run `make test`. Dependencies use version ranges; this is not a fully locked build. Record `git rev-parse HEAD`, installed package versions and provider model IDs when reproducing. Tests do not require a GPU.

8. Next steps

Recruit a named inventor/R&D pilot; define ten patent tasks with reviewed expected evidence. Compare the source-only baseline with the model reading, and 8B with 70B, measuring usefulness, error rate, latency and tokens. Recheck all 14 drawing pages with a human reader. Run cold/warm and ten-concurrent-user checks. Validate a local inference setup and resolve dependency/data rights before wider distribution.

The confirmed team is Rafael Gross and Ricardo Gross, both professionals. The entrant's GitHub identity and Devpost email are still required for the form.

The public demo is an edited walkthrough of genuine application captures, with synthetic English narration, lasting 111 seconds. It shows live EPO search, English and Bernese readings, Plus, drawing labels, source fallback and visible model activity. It does not validate the private workspace workflow. The report, repository and video are prepared; final terms acceptance and submission remain pending. Dependency obligations are disclosed in section 4 and the dependency inventory.

License

RGross-authored report: Creative Commons Attribution 4.0 (CC-BY-4.0). External material retains its own terms.

References

- **R1:** EPO Technology Dashboard 2025.
- **R2:** Apertus: openness and project background.
- **R3:** Apertus values and FAQ.
- **R4:** Official Track 2B template and report, reviewed 16 October 2026; template revision `09f8e9e2b588db3325869b23c6d5132f5c3756f7`.
- **R5:** APatent evaluation record.
- **R6:** Getting Started Guide and submission instructions, reviewed 16 October 2026.
- **R7:** PyMuPDF licensing.
- **R8:** Event terms, section 6.